serial communication. The Raspberry Pi software then processed the incoming data stream in real time to generate GUI corrective prompts, calculate session statistics, export CSV logs, and transmit motor control commands back to the Arduino. Integrated verification confirmed stable bidirectional communication and synchronization between hardware and software subsystems throughout repeated CPR trials.

5.7.2 Quantitative Integrated Verification Data

Extensive integrated verification testing was conducted using exported Raspberry Pi session logs generated during repeated CPR trials. A total of more than 150 CPR session CSV files were generated across multiple users and operating conditions throughout integrated system verification. These exported logs contained over 42,000 recorded sensor and compression data rows representing approximately 75 minutes of cumulative CPR session activity. Review of the exported data confirmed stable timestamp progression, proper CSV formatting, and successful logging of all required CPR performance metrics, including compression depth, recoil, compression rate, hand placement centroid position, applied force, and motor activation behavior.

The exported session logs demonstrated successful real-time acquisition and synchronization of multiple sensing modalities throughout active CPR operation. Compression depth measurements collected from the time-of-flight sensor consistently reflected dynamic user compression behavior during testing. Across representative sessions shown in Figures 5.6.2.4–5.6.2.6, measured compression depths frequently operated within the programmed target range of 50–60 mm, while also successfully capturing intentional under-compression and over-compression test conditions.

Compression rate measurements similarly demonstrated successful detection of changing CPR pacing conditions throughout active sessions. The exported data showed that the system correctly tracked rates both inside and outside the programmed 100–120 compressions-per-minute target region. Variations in compression pacing generated corresponding GUI prompts (“FASTER” and “SLOWER”), demonstrating successful integration between sensor acquisition, software analysis, and real-time user feedback systems.

Load cell force measurements exhibited stable periodic behavior synchronized with active chest compressions throughout verification testing. Representative force plots shown in Figures 5.6.2.4–5.6.2.6 demonstrate consistent pulsed force responses during active compressions, with measured forces frequently exceeding 200 N during deeper compression events. The data additionally confirmed successful detection of recoil behavior and lean conditions during testing.

The FSR hand placement subsystem also demonstrated stable integrated operation during repeated CPR sessions. Exported centroid tracking data confirmed continuous monitoring of hand placement deviations along both the horizontal and vertical axes of the manikin surface. Figures 5.6.2.4–5.6.2.6 demonstrate successful detection of intentional hand placement offsets exceeding the programmed ±15 mm target region, which generated corresponding directional GUI prompts (“MOVE UP,” “MOVE DOWN,” “MOVE LEFT,” and “MOVE RIGHT”). Verification testing confirmed that the FSR subsystem successfully interacted with the Raspberry Pi software to provide practical real-time hand placement correction feedback throughout active sessions.

5.7.3 Integrated Feedback and Motor Verification

Integrated testing additionally verified synchronization between CPR quality evaluation and the blood flow visualization subsystem. During active sessions, compression events satisfying the programmed depth and recoil thresholds triggered short-duration motor pulses which produced visible fluid movement through the tubing system. Non-qualifying compressions did not activate the motor subsystem. This behavior demonstrated successful closed-loop integration between the sensing hardware, Arduino processing logic, Raspberry Pi evaluation software, and physical feedback subsystem.

At session completion, the Raspberry Pi software evaluated overall CPR performance using averaged compression metrics derived from the exported session data. Successful sessions generated pass conditions which triggered sustained post-session motor activation commands transmitted back to the Arduino. Verification testing confirmed stable synchronization between session scoring behavior, GUI result presentation, and physical blood flow visualization responses.

5.7.4 Integrated Verification Results

The SMART CPR training device successfully demonstrated stable integrated operation throughout extensive verification testing. Repeated trials confirmed reliable interaction between sensing hardware, embedded processing systems, serial communication infrastructure, Raspberry Pi software analysis, GUI feedback systems, session logging functionality, and the blood flow visualization subsystem. No corrupted export files, communication lockups, unintended Arduino resets, or GUI crashes were observed throughout integrated testing.

Overall, the quantitative verification data demonstrated that the SMART CPR system successfully integrated real-time sensing hardware, software analysis, graphical feedback, data collection, and physical blood flow visualization into a unified CPR training platform capable of evaluating and reinforcing proper CPR technique under extended repeated operation.

5.8 Verification Limitations

Regarding the limitations encountered during verification of the Smart CPR Training Device, the team was constrained by the testing systems and laboratory resources available for use. The MTS machine was identified as a temporary testing option through one team member's affiliation with a research lab, but access to this equipment was not guaranteed or formally committed. As a result of limited laboratory access, the team relied on alternative verification methods, including first-order mathematical analysis, spatial measurement principles, calibrated weights, video recording, cross-correlation analysis, and visual inspection. The University of Arizona High Performance Computing system was also used to support verification of the compression module through computational analysis. These methods allowed the team to continue evaluating system performance despite limited access to formal testing equipment.